

Hydrodynamic focusing

X21 (2021) — English translation

Some applications require extremely narrow fluid flows. The narrowness of the flow significantly accelerates the mixing of the substance with the environment, which makes it possible to study fast chemical reactions. Also, the use of narrow flows allows you to reduce the amount of substance used. If you create a narrow flow using a piston that forces fluid through a narrow opening, the resulting flow is extremely turbulent. This makes experiments very difficult. Making fairly narrow channels is also quite labor-intensive. These difficulties can be circumvented by compressing the flow by surrounding it with a flow of another fluid. Depending on the flow parameters, the width of the central part of the flow can significantly decrease. This phenomenon is called hydrodynamic focusing. In this problem we will consider the simplest models of hydrodynamic focusing.

Part A. Flow of a single fluid (2.7 points)

It is known that if the fluid velocity depends only on the coordinate x and is directed along the axis y , then the force

$$F = \eta S \frac{dv}{dx}.$$

S – площадь соприкосновения жидкостей. Сила, действующая на более медленно движущийся слой направлена в сторону его движения. Рассмотрим течение жидкости в щели между двумя параллельными плоскостями. Расстояние между двумя плоскостями d , ширина плоскостей w , длина L . Разность давлений на краях системы Δ acts between two layers of fluid P . Вязкость жидкости η . Ось x направлена перпендикулярно плоскостям, ось y – в направлении течения жидкости. Мы считаем, что $d \ll w$, поэтому можно пренебрегать эффектами на границе считать, что скорость течения зависит только от x и всюду направлена вдоль оси y . We consider the flow to be stationary everywhere, that is, the velocity distribution does not depend on time.

A1 Let $v(x)$ be the velocity of fluid flow in the slot. The origin of the coordinates of the Ox axis is located in the middle between the planes, and the planes limiting the flow are located at $x = \pm d/2$. The pressure difference at the opposite edges of the slit is ΔP . Considering the forces acting on an infinitesimal layer of liquid of thickness dx , obtain a differential equation for $v(x)$. Find $v(x)$ from it. On the walls of the slit, the speed becomes 0. Find the average speed of fluid flow $\langle v \rangle$ (express it through ΔP , L , η , d). Find the flux (flux is the volume of liquid flowing through the slit per unit time) of the liquid through the slit. Express your answer using ΔP , L , η , d , w . **1.00**

A2 Now consider the flow of liquid through a long circular tube of length L and radius r . For reasons of symmetry, the speed of flow through a tube depends only on the distance to its axis. At the boundary of the tube, the speed becomes 0. Find the dependence of the speed on the distance to the ρ axis. The pressure difference between the ends of the tube is ΔP , the viscosity of the liquid is η . Find the volumetric flow rate q as the liquid flows through the tube. Express your answer using ΔP , L , r , η . **0.80**

For both flow through a slit and flow through a tube, the flow is proportional to the pressure difference at the ends. The dependence of flow on pressure can be written in the form $q = \Delta P/R$, where R is the hydrodynamic resistance of the tube (or slot).

A3 Find the hydrodynamic resistance of the slot R_s and tube R_t . Express your answer in terms of the geometric dimensions given in paragraphs A1 and A2, respectively, and in terms of the fluid viscosity η . **0.20**

A4 A liquid of viscosity η flows through a tube of radius r and length L under the influence of a pressure difference ΔP (as in A2). Find the power of viscous friction forces. **0.70**

Part B. Two types of liquids (1.4 points)

Now let two types of liquids flow between the planes. The sample we are interested in flows in the central section (we will denote all quantities related to it with index 1). The width of its flow is w_1 . The remaining part of the flow (at the edges) is occupied by liquid 2, which compresses the sample (see figure). In subsequent parts of the problem, we will be interested in the ratio of the sample width to the total flow width w_1/w . We will assume that the flow width is much larger than the slot thickness $w_1 \gg d$. Then the influence of the area of contact of liquids on the flow can be neglected, and it can be assumed that the speed of the liquid at each point depends only on its properties and the x coordinate (the Ox axis is perpendicular to the planes). At the boundaries of the gap at $x = \pm d/2$ the velocity becomes 0 as in part A.

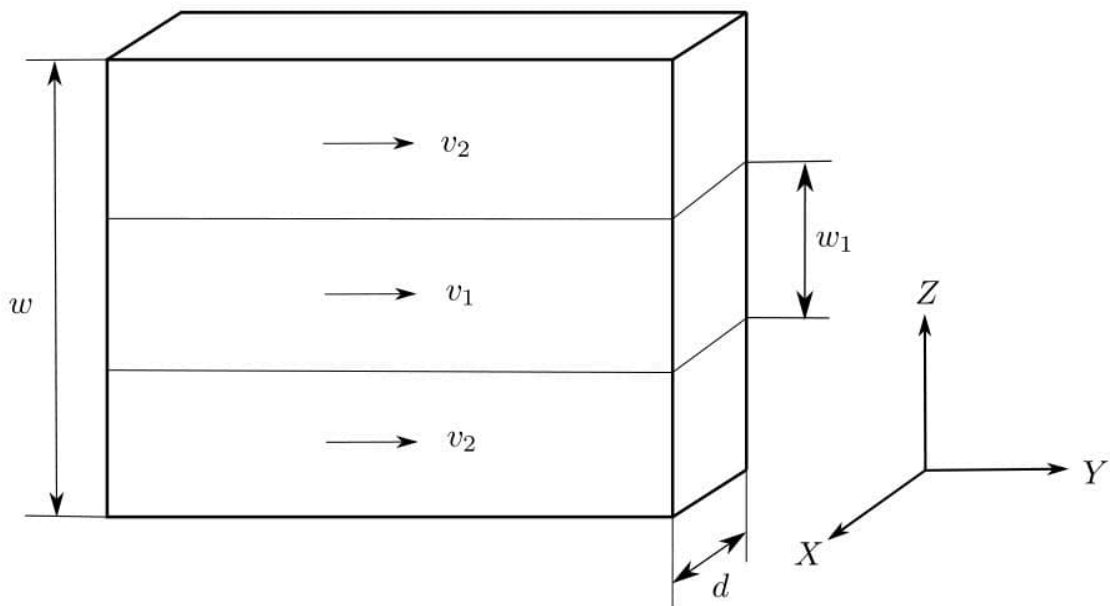


Fig. 1.

B1 Let the volumetric flow rates q_1 and q_2 of liquids 1 and 2, as well as their average velocities $\langle v_1 \rangle$ and $\langle v_2 \rangle$, be given. Find the relative flow width of sample w_1/w . **0.40**

- B2** Let the viscosities of the liquids η_1 and η_2 be given, and their average velocities are determined by the results of part A. Express w_1/w in terms of q_1 , q_2 , η_1 and η_2 . Effects that occur at the interface between fluids 1 and 2 can be neglected, since they are only important in a region with a width of the order of $d \ll w$. **0.40**

Liquids 1 and 2 come from two reservoirs located at heights h_1 and h_2 under the influence of gravity. These tanks are connected to the main part of the installation by a system of pipes. We can assume that the fluid flow through each tube does not depend on time and is proportional to the pressure difference at its ends

$$q_i = \Delta P_i / R_i.$$

Здесь R_i is the hydrostatic resistance of the tube. This resistance depends on the parameters of the tube and the properties of the liquid. The pressure difference at the ends of the tube is created by gravity.

In order to make the flow independent of the slot parameters, it is necessary that the resistance of the tubes be much greater than the resistance of all other parts of the circuit. To do this, special very narrow cylindrical capillaries - rheostats - are added to the tubes. In what follows, we will assume that the entire resistance of the supply tubes is determined by the resistance of these rheostats. Let us assume that the geometric dimensions of the rheostats (their radii and lengths) for the sources of both liquids are the same.

- B3** Find the flow ratio q_1/q_2 if the densities of the liquids (ρ_1 and ρ_2), their viscosities (η_1 and η_2) and the heights to which the reservoirs h_1 and h_2 are raised are given. **0.30**

- B4** Find the corresponding ratio of stream widths w_1/w . **0.30**

Part C. Considering gravity (2.8 points)

Experiments show that in the case of different densities of fluids 1 and 2, the focused flow is most stable when the flow direction is vertical. In this case, the width of the flow may change under the influence of gravity.

- C1** Let us again consider the flow of one fluid between planes. However, now the speed is created not only due to the pressure difference, but also due to gravity. The geometry is the same as in part A1, the force of gravity is directed along the y axis. The fluid flow is aligned with the force of gravity. Find the average speed of the fluid. Express your answer in terms of ρ , g , η , d , ΔP , L . **0.40**

- C2** Let now two types of fluids flow between the planes, as in part B. Having written expressions for the average speeds of liquids 1 and 2, exclude the pressure difference from them (it is the same for both liquids) and find the ratio $\langle v_1 \rangle / \langle v_2 \rangle$. Express your answer using $\langle v_2 \rangle$, η_1 , η_2 , d , g , ρ_1 , ρ_2 . **0.80**

- C3** Substitute the expression for the average velocities into the formula for the relative width of the focused flow from B1. Obtain an equation for determining the width of the form **1.00**

$$\frac{w}{w_1} = \dots$$

В правую часть могут входить отношение $\frac{w_1}{w}$ и безразмерные параметры $\alpha = q_2\eta_2/q_1\eta_1$, $\beta = (\rho_1 - \rho_2)q_G/\rho_2q_2$. Здесь $q_G = \rho_2gd^3w/12\eta_2$ is a quantity with the dimension of fluid flow, characterizing the influence of gravity.

- C4** Solve this equation and find the relative width of the focused flow w_1/w . Express your answer in terms of dimensionless quantities α, β . **0.60**

Part D. Experiment (3.1 points)

In an experiment, it is possible to directly regulate not the amount of liquid flow, but the height at which the corresponding reservoir is located. Gravity acceleration $g = 9.8 \text{ m/s}^2$. Water is used as liquid 2, which is constantly supplied from height $h_2 = 0.5 \text{ м}$. The density of water is $\rho_2 = 10^3 \text{ kg/m}^3$. Water viscosity $\eta = 8.9 \cdot 10^{-4} \text{ kg/(m} \cdot \text{с)}$ The rheostat is a capillary with radius $r = 130 \text{ мкм}$ and length $L = 0.5 \text{ м}$. A slit with length $l = 43 \text{ мм}$, width $w = 3 \text{ мм}$, $d = 175 \text{ мкм}$ is used.

- D1** Find the hydrodynamic resistance of each of the rheostats if water flows through them. Let a salt solution be used as liquid 1, the density of which is $\rho_1 = 1.15 \cdot 10^3 \text{ kg/m}^3$, and the viscosity coincides with the viscosity of water. Find q_G for such a gap. **0.60**

- D2** Find the relative width of the flow of liquid 1 (salt solution) w_1/w for cases when liquid 1 is supplied from heights $h_1 = 0.25 \text{ м}$, 0.5 м , 1.0 м . **0.30**

- D3** How will the value of the flow width change if we neglect the influence of gravity? Calculate the relative flow width for the same parameters $h_1 = 0.25 \text{ м}$, 0.5 м , 1.0 м as in the previous paragraph, without taking into account gravity. **0.30**

- D4** Let now some liquid 1 flow through the center, the density of which is equal to $\rho'_1 = 0.8 \cdot 10^3 \text{ kg/m}^3$, and the viscosity coincides with the viscosity of water. Find the ratio w_1/w for the values $h_1 = 0.25 \text{ м}$, 0.5 м , 1.0 м . **0.30**

- D5** Plot qualitatively on one graph the dependence of w_1/w on h_2/h_1 for three cases: a) there is no gravity, b) a salt solution flows in the center, c) a liquid with density ρ'_1 flows in the center. All parameters are the same as in the previous paragraphs, the height h_2 is fixed, only h_1 changes. **0.80**

- D6** In all calculations we assumed that the flows were independent of time. Let the water tank (liquid 2) be a cylinder of radius $r_r = 20 \text{ мм}$. Estimate the time it takes for the water level in the tank to decrease by $\delta h_2 = 10 \text{ см}$. If the experiment is carried out over a period of this order, a change in the water level in the tank will have virtually no effect on it. The starting level is $h_2 = 0.5 \text{ м}$. **0.80**

Source: <https://pho.rs/p/105>